

CHE 435/525 *Process Systems Analysis and Control*

Lecture 10: **Zeros, Inverse Response, and Non-Invertibility**

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Recapture: Delayed systems and higher-order systems

- First-order plus time delay: $G(s) = \frac{K}{\tau s + 1} e^{-\theta s}$
 - ▶ How does time delay arise in a process? Lack of mixing (plug flow)
 - ▶ How does its step response look like?
- Higher-order system:

$$G(s) = \frac{B(s)}{A(s)} = \frac{b_m s^m + b_{m-1} s^{m-1} + \cdots + b_1 s + b_0}{a_n s^n + a_{n-1} s^{n-1} + \cdots + a_1 s + a_0}$$

- ▶ Stable if all poles are on LHP
- ▶ If stable, $G(0) = K$ is the gain

Learning objectives of this lecture

- ① Explain the implications of **right-half-plane zeros**
- ② Convert transfer functions into **gain/time constant form**
- ③ Recognize the form of transfer function based on step response plot

Zeros

- **Recall:** Poles are roots to denominator $A(s)$. Their locations on the complex plane determine stability.
- **Question:** Zeros are roots to numerator $B(s)$. What do their locations imply?
- To best investigate this, let's always treat the TF with a factorization!

$$G(s) = \frac{B(s)}{A(s)} = \frac{b_m(s - z_1) \cdots (s - z_m)}{a_n(s - p_1) \cdots (s - p_n)}$$

- ▶ The poles and zeros have units [1/time]
- ▶ Denote “denominator time constants” $\tau_1 = -1/p_1, \dots, \tau_n = -1/p_n$, then $(s - p_1) \cdots (s - p_n) = \text{const} \cdot (\tau_1 s + 1) \cdots (\tau_n s + 1)$.
- ▶ Denote “numerator time constants” $\bar{\tau}_1 = -1/z_1, \dots, \bar{\tau}_m = -1/z_m$, then $(s - z_1) \cdots (s - z_m) = \text{const} \cdot (\bar{\tau}_1 s + 1) \cdots (\bar{\tau}_m s + 1)$.

Gain/time constant form

- Thus, the TF can be converted to a standard **gain/time constant form**

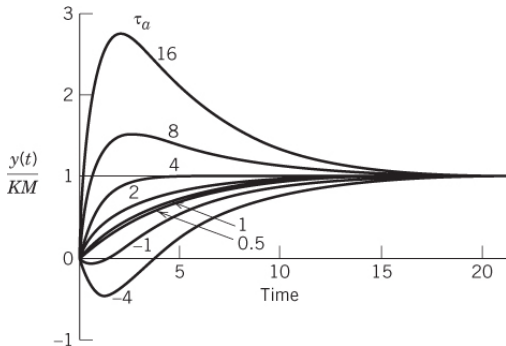
$$G(s) = K \frac{(\bar{\tau}_1 s + 1) \dots (\bar{\tau}_m s + 1)}{(\tau_1 s + 1) \dots (\tau_n s + 1)}$$

- ▶ $K = G(0)$ is the gain
- ▶ $\tau_j = -1/p_j$. **If all τ_j are on RHP, then the system is stable.**
- ▶ $\bar{\tau}_j = -1/z_j$. **Question:** What do the locations of numerator time constants imply?
- Practice: Convert $G(s) = \frac{s+4}{s^2+6s+5}$ into gain/time constant form
- Practice: Convert $G(s) = \frac{-4s+0.5}{s^2+6s+5}$ into gain/time constant form

Inverse response

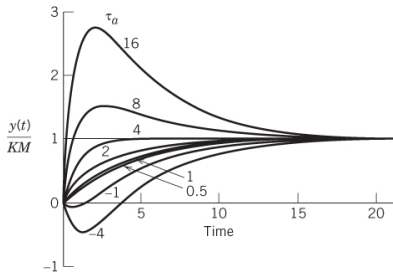
- When a numerator time constant is **negative**, namely when there exists a **right-half-plane zero**, there can be **inverse response** – “sign-flipping”

$$G(s) = \frac{K(\tau_a s + 1)}{(\tau_1 s + 1)(\tau_2 s + 1)} \quad : \quad \tau_1 = 4, \tau_2 = 1$$



- Distinguish **inverse response** / **overshoot** / **oscillations**!

Inverse response: Interpretation



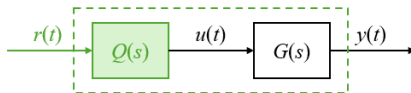
- Two first-order dynamics in competition (for simplicity, $K = 1$):

$$G(s) = \frac{\tau_1 - \tau_a}{\tau_1 - \tau_2} \cdot \frac{1}{\tau_1 s + 1} + \frac{\tau_a - \tau_2}{\tau_1 - \tau_2} \cdot \frac{1}{\tau_2 s + 1} = \frac{k_1}{\tau_1 s + 1} + \frac{k_2}{\tau_2 s + 1}$$

- ▶ The first dynamics is slow (τ_1); the second dynamics is fast (τ_2)
- ▶ When $\tau_a < 0$, we have $k_1 > 1$, $-1 < k_2 < 0$. The first gain is larger in absolute value and **ultimately dominates**, but the second dynamics exhibits itself **transiently**
- ▶ $y'(0) = \lim_{s \rightarrow \infty} G(s) = 0$, $y''(0) = \lim_{s \rightarrow \infty} sG(s) = K\tau_a/\tau_1\tau_2 < 0$

RHP zeros: Non-invertibility

- **Imagine:** Given the plant $G(s)$ with inverse response, you want to design connect the plant to an extra device $Q(s)$, so that $G(s)Q(s)$ has no inverse response.



- ▶ If so, you manipulate $r(t)$ (the input to $Q(s)$), so that the output of $Q(s)$ will “automatically” manipulate the plant. The overall system $r \rightarrow y$ has no inverse response. [Useful later in process control.]
- When $G(s)$ has a RHP zero, to **cancel** this RHP zero, $Q(s)$ must have a corresponding RHP pole (i.e., $Q(s)$ is unstable)
- In reality, you can't **exactly “assign” a pole**. The resulting $G(s)Q(s)$ will contain both a RHP pole and a RHP zero ... even worse!
- **Conclusion:** RHP zeros are **non-invertible / irremovable**.
- The same holds for time delay $e^{-\theta s}$.

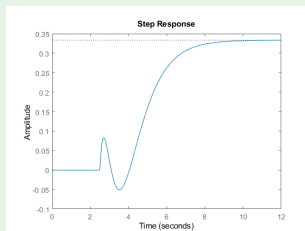
Summary: Time-domain analysis

- **Step, impulse (,ramp) responses**
- Characteristics of the step response of **first- and second-order systems**
- **Poles, zeros, gain, time delay** and their implications
- **Standard forms** of transfer functions and their conversions

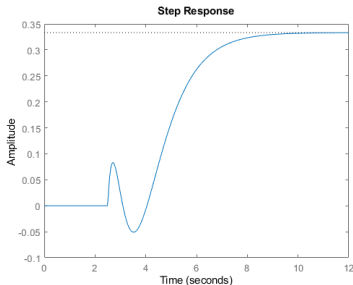
Exercise: Recognizing the form of a TF model

Step response of a complicated process

The engineer used a unit step input and got the following step response. Determine the **structure** of $G(s)$ (not its parameters).



Exercise: Recognizing the form of a TF model

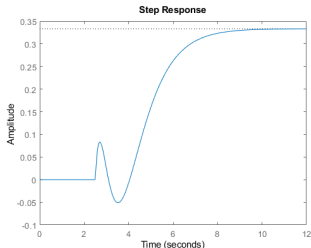


- Start with the standard gain/time constant form.

$$G(s) = K \frac{(\bar{\tau}_1 s + 1) \dots (\bar{\tau}_m s + 1)}{(\tau_1 s + 1) \dots (\tau_n s + 1)} e^{-\theta s}$$

- 1 Poles: Is it stable? Is it oscillatory?
- 2 Gain & Delay: What is the ultimate output/input ratio? Deadtime before response?
- 3 Zeros: Is there inverse response or overshoot? How many RHP zeros?

Exercise: Recognizing the form of a TF model



- ① Poles: Negative real (stable system, non-oscillatory).
 - ▶ Denominator: $(\tau_1 s + 1) \cdots (\tau_n s + 1)$, all $\tau_i > 0$.
- ② Gain: $K \approx 0.33$; Delay: $\theta \approx 2.5$
- ③ Zeros: Two on RHP (inverse response occurs twice, no overshoot).
 - ▶ Numerator: $(\bar{\tau}_1 s + 1)(\bar{\tau}_2 s + 1)$, both $\bar{\tau}_i$ on LHP. Also, $n \geq 3$ since $m = 2$.
- ④ Actual TF:

$$G(s) = \frac{1}{3} \cdot \frac{\frac{1}{2}s^2 - s + 1}{(s + 1)(\frac{1}{2}s + 1)(\frac{1}{3}s + 1)} e^{-2.5s}.$$